Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series

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Abstract

[Abstract: write last.]

Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, deep learning

1. Introduction

The Global Navigation Satellite System (GNSS) enables precise positioning with up to 1 millimetres in the horizontal and 3-5 and 3-5 millimetre in the vertical axes.

[Paragraph 2: common-mode error and the filters that remove it.]
[Paragraph 3: why this criterion is a problem.]
[Paragraph 4: Bachelot et al. and what remains open.]
[Paragraph 5: what this study does.]

2. Data and processing

[Section 2, paragraph 1: overview of the processing chain (Fig. 1).]

2.1. Data

[Paragraph 2: what one daily solution is.]
[Paragraph 3: station selection and the network (Fig. 2a).]

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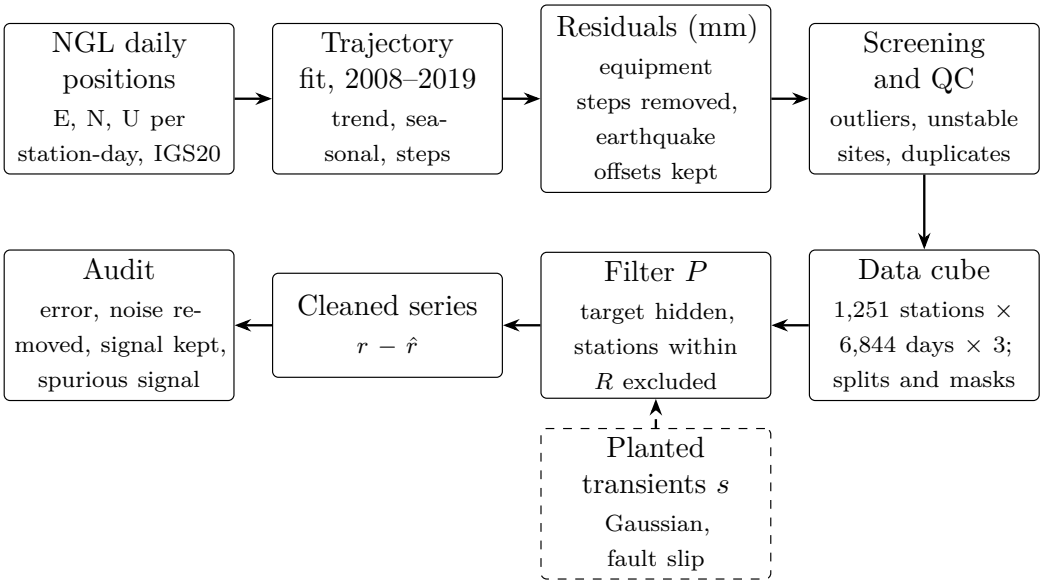


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit. [Check caption.]

2.2. From positions to residuals

[Paragraph 4: the trajectory model and why it is fitted on the training years only.]

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[s_k \sin(2\pi k t) + c_k \cos(2\pi k t) \right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t) \tag{1}$$

[Paragraph 5: equipment steps removed, earthquake offsets kept (Fig. 3b).]
[Paragraph 6: outlier screening and quality control.]
[Paragraph 7: the data cube, splits and held-out stations.]
[Paragraph 8: why we filter displacement rather than rates.]

3. The signal-loss audit

3.1. Leave-one-out filtering and the exclusion radius

[Paragraph 1: leave-one-out, the exclusion radius R and leak checks.]

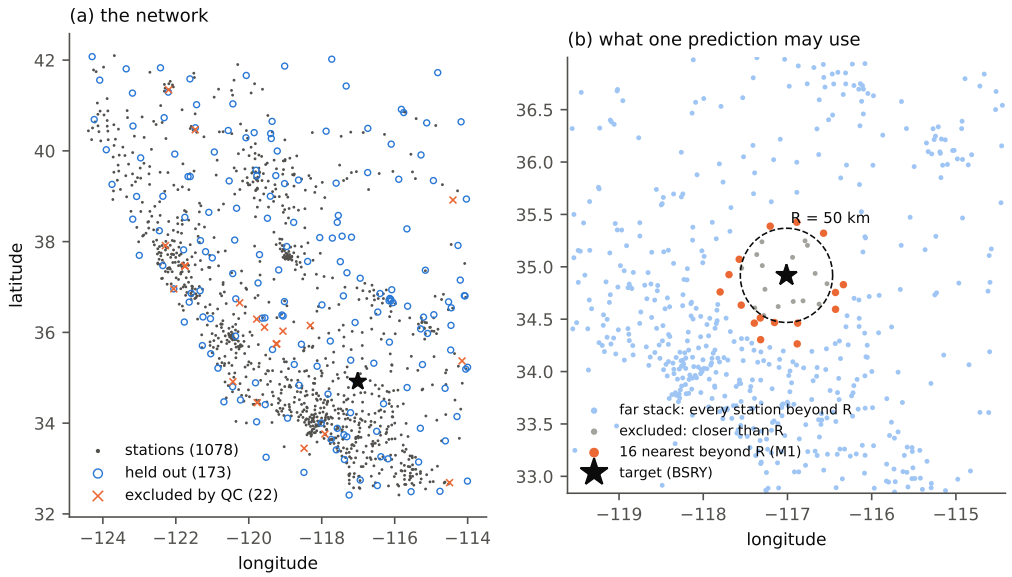


Figure 2: (a) Stations used, held out and excluded. (b) What a prediction for one target may use at $R = 50$ km. [Expand caption.]

3.2. Four measures

[Paragraph 2: reconstruction error.]
[Paragraph 3: noise removed.]
[Paragraph 4: signal kept.]

$$\rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \qquad \Delta = \Big(r + s - P(r+s)\Big) - \Big(r - P(r)\Big) \tag{2}$$

[Paragraph 5: spurious signal.]

3.3. Planted and real signals

[Paragraph 6: Gaussian transients.]
[Paragraph 7: fault-slip events from the event library.]
[Paragraph 8: real earthquake offsets and Ridgecrest.]

3.4. Protocol

[Paragraph 9: decisions, freeze date, bootstrap, seeds.]

Table 1: Filters audited. [Check descriptions.]

Name	Description	Parameters
Target-inclusive		
C1	Mean of the whole network, target included	0
C3	Sub-regional probabilistic PCA; 1, 4 or 8 regions × 1–3 components	∼3.7k/comp.
M2-self	Robust weighted median of the 16 nearest and the target	2
Distance-excluded		
Far stack	Mean of every station beyond R; D1 at $R = 400$ km	0
Leave-one-out over nearest neighbours		
M1	Distance-weighted mean of the 16 (64, 256) nearest beyond R	1
M2	Robust weighted median of the 16 nearest beyond R	2
M3a / M3b	Ridge regression on 16 neighbours, per station / pooled	65/station; 257
M4a / M4b	M3 per time band (under 7, 7–60, 60–400, over 400 days)	4 × M3
Learned		
M3k	Learned kernel: neighbour weight from distance, azimuth and noise level; 16 or 256 neighbours	1,411
M5	Space–time attention: 32 neighbours and the far stack, 64-day window	212,048
M5-aug	M5 trained with planted transients in its inputs	212,048

4. Filters

[Paragraph 1: the three families.]
[Paragraph 2: classical and regression filters.]
[Paragraph 3: learned models.]

5. Results

5.1. The trade-off between noise removed and signal kept

[Paragraph 1: the overall trade-off (Fig. 5).]
[Paragraph 2: what the radius does.]

Table 2: Main results at $R = 0$ unless stated. *Validation years; to be replaced by test-year results after the method freeze.* Error: fast leave-one-out error (lower is better; zero prediction 0.723). Kept: median retention of 25- and 100-km Gaussian transients; afterslip at 15–40 km from the patch; real earthquake offsets.

Filter	Error	Noise removed	Kept 25 / 100 km	Spurious	Afterslip	Real offsets
C1	–	26.7%	0.99 / 0.83	0.11	1.00	0.98
C3 (9 settings)	–	25.1–26.4%	0.88–0.99 / 0.30–0.84	0.27–1.52	0.97–1.00	0.90–0.95
M2-self	–	25.0%	0.24 / 0.00	0.10	0.19	0.60
Far stack	0.509	26.6%	0.99 / 0.83	0.11	1.00	0.98
D1 ($R = 400$ km)	0.526	24.9%	1.00 / 1.00	0.24	1.00	0.99
M1	0.531	22.2%	0.26 / 0.00	0.02	0.09	0.72
M1, 256	0.507	26.4%	0.84 / 0.21	0.10	0.96	0.83
M2	0.558	18.4%	0.27 / 0.00	0.11	0.23	0.69
M3a	0.527	22.6%	0.43 / 0.17	0.04	0.22	0.77
M3b	0.534	21.4%	0.30 / 0.07	0.04	0.24	0.74
M4a	0.517	24.9%	0.56 / 0.37	0.03	0.53	0.75
M4b	0.532	21.5%	0.33 / 0.11	0.04	0.31	0.72
M3k	0.517	23.9%	0.28 / 0.01	0.03	0.11	0.81
M3k, 256	0.499	27.1%	0.71 / 0.22	0.18	0.73	0.88
M5	0.504	26.8%	0.80 / 0.56	0.15	0.87	0.88
M5-aug	0.508	26.5%	0.97 / 0.80	0.17	0.99	0.98

5.2. Number of stations versus distance

[Paragraph 3: count beats distance (Fig. 6).]

[Paragraph 3b: regression on the nearest stations (M3, M4).]

5.3. Realistic transients

[Paragraph 4: near field kept, far field removed (Fig. 7).]

[Paragraph 5: shallow slow slip and the spurious signal it causes.]

5.4. Real earthquake offsets

[Paragraph 6: real offsets confirm the planted results (Fig. 8).]

5.5. Learned models

[Paragraph 7: the learned kernel.]

[Paragraph 8: more neighbours, more noise removed, less signal kept.]

[Paragraph 9: the attention network with and without planted-transient training.]

6. Discussion

[Paragraph 1: what practitioners should do.]
[Paragraph 2: why local filters remove the far field and why it matters.]
[Paragraph 3: relation to Bachelot et al.]
[Paragraph 4: what this means for deep learning.]
[Paragraph 5: limitations and future work.]

7. Conclusions

[Conclusions: write after the discussion.]

Data and code availability

[Data and code availability.]

Declaration of generative AI in the writing process

[AI declaration: write at the end, describing what was actually done.]

References

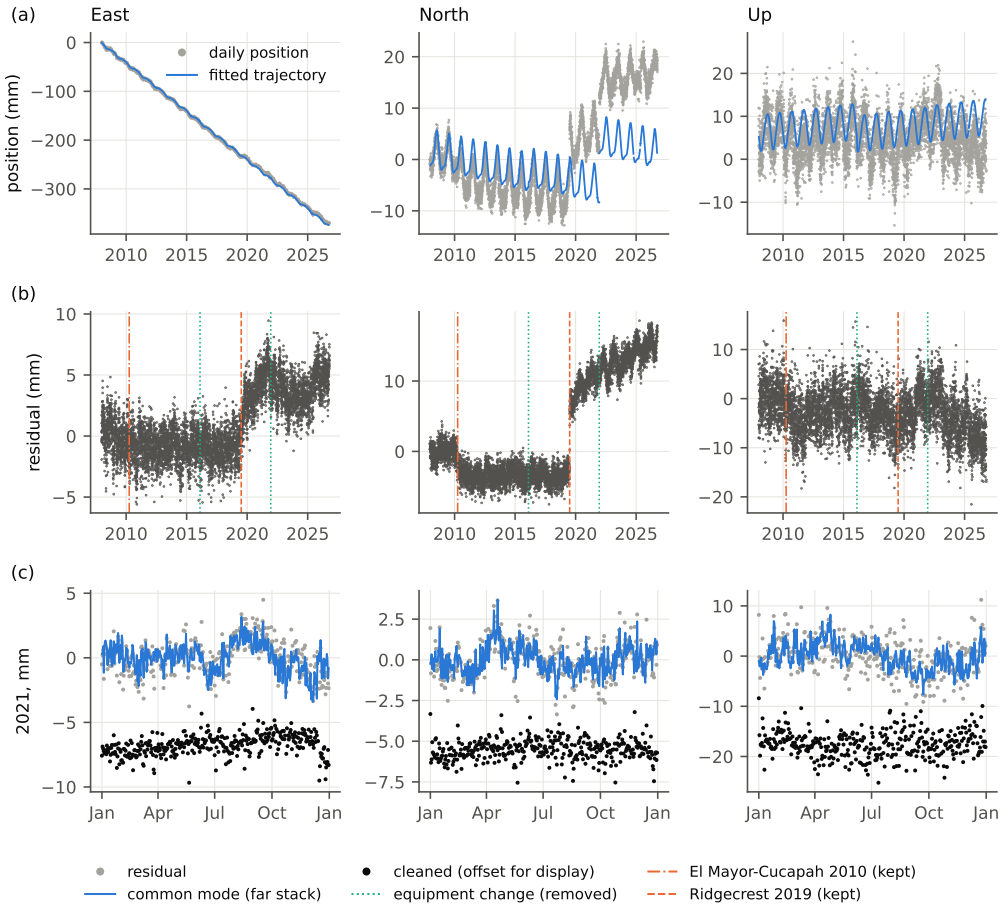


Figure 3: From positions to residuals at station BSRY. [Expand caption: (a) positions and trajectory, (b) residual, (c) one validation year with the network common mode and the cleaned series.]

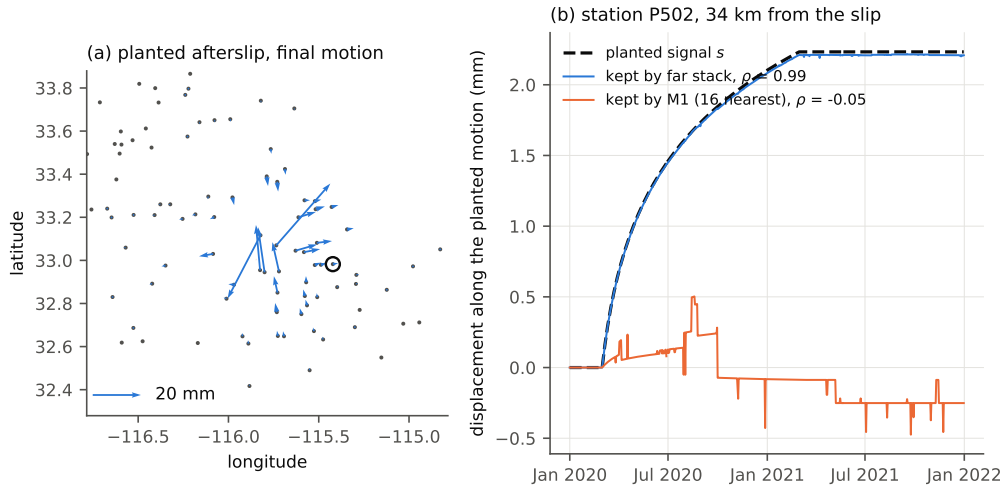


Figure 4: Measuring signal kept. [Expand caption; note that P502 is the median station among 17 at 15–40 km.]

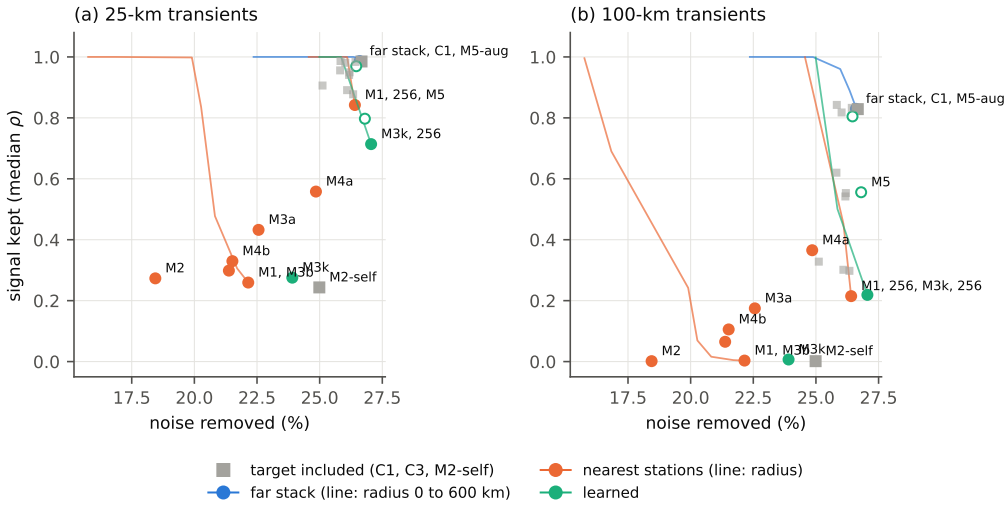


Figure 5: [Caption.]

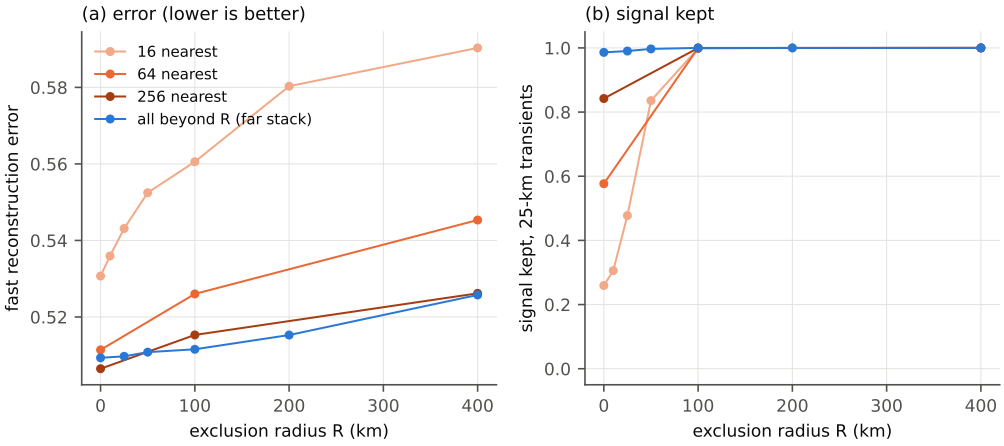


Figure 6: [Caption.]

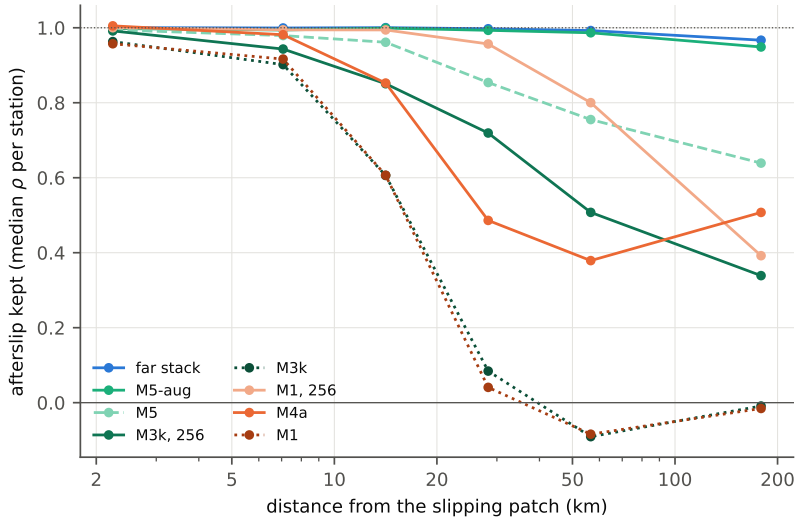


Figure 7: [Caption.]

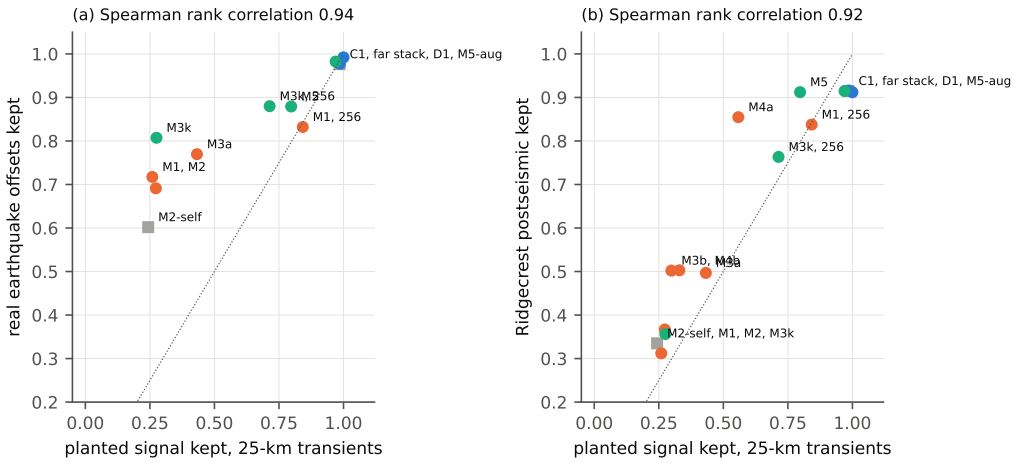


Figure 8: [Caption.]